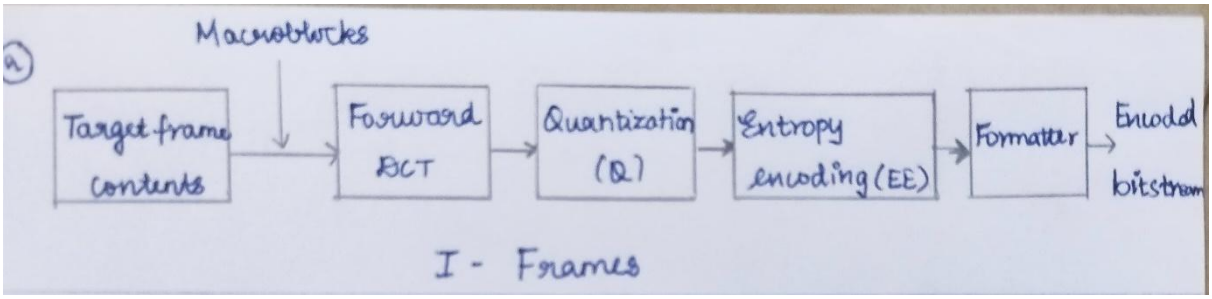
KONGU ENGINEERING COLLEGE, PERUNDURAI - 638 060

SEMESTER ODD|CONTINUOUS ASSESSMENT TEST – III

PART - A (10 × 2 = 20 Marks)			
ANSWER ALL THE QUESTIONS			
1.	Voiced sound: These are generated through vocal chords. e.g letters m,v,and i. Unvoiced sound**:** With these vocal chords are open. e.g letters f and s.	CO4	K1
2.	A pre-computed set of templates are held by the encoder and decoder is known as a template codebook.	CO4	K1
3.	1.Sampling 2.Quantization 3.Coding	CO4	K1
4.	Pitch, Period and Loudness	CO4	K1
5.	**Frequency Masking:** When multiple signals are present in audio, a strong signal may reduce the level of sensitivity of the ear to other signals which are near to it in frequency. **Temporal masking:** When the ear hears a loud sound it takes a short but a finite time before it could hear a quieter sound.	CO4	K1
6.	**Spatial redundancy:** Correlation between adjacent pixels of same frame. **Temporal redundancy:** Correlation between adjacent frames.	CO5	K1
7.	Essential units involved in encoding of I frames. 	CO5	K1
8.	1. Use of fixed-length video packets as the base-level data structure instead of GOBS; 2. New Variable Length Coding (VLC) scheme based on reversible VLCs.	CO5	K1
9.	AVO- Audio Visual Object BIFS-BInary Format for Scenes	CO5	K1
10.	The compression algorithm used with MPEG-1 differs from that used in the H.261 standard: 1. Time-stamps can be inserted within a frame to enable the decoder to resynchronize more quickly in the event of one or more corrupted or missing macroblocks. The number of macroblocks between two timestamps is known as a slice and a slice equal to 22 macroblocks in a line. 2. Introduction of B-frames which increases the time interval between I and P frames. To allow for the resulting increase in the separation of moving objects with P-frames, the search window in the reference frame is increased.	CO5	K1
Part – B (3 × 10 = 30 Marks)			
ANSWER ANY THREE QUESTIONS			
11.	**Differential Pulse Code Modulation:** □ Differential pulse code modulation is a derivative of the standard PCM □ It uses the fact that the range of differences in amplitudes between successive samples of the audio waveform is less than the range of the actual sample amplitudes □ Hence fewerbits are required to represent the difference signals than in case of PCM for the same sampling rate. It reduces the bit rate requirements from 64kbps to 56kbps. **Adaptive Differential Pulse Code Modulation :** ➢ Better sound quality at the cost of added complexity. ➢ Input speech BW is extended from 50-7kHz compared with 3.4kHz for a standard PCM system ➢ Wider BW give rise to high quality. This is a need in video conferencing. ➢ This uses Sub band coding ➢ In this coding input signal prior to sampling is passed through two filters:	(10)	CO4	K2

- One passes only signal frequencies in the range 50Hz - 3.5kHz
- Other passes only frequencies in the range 3.5kHz - 7kHz

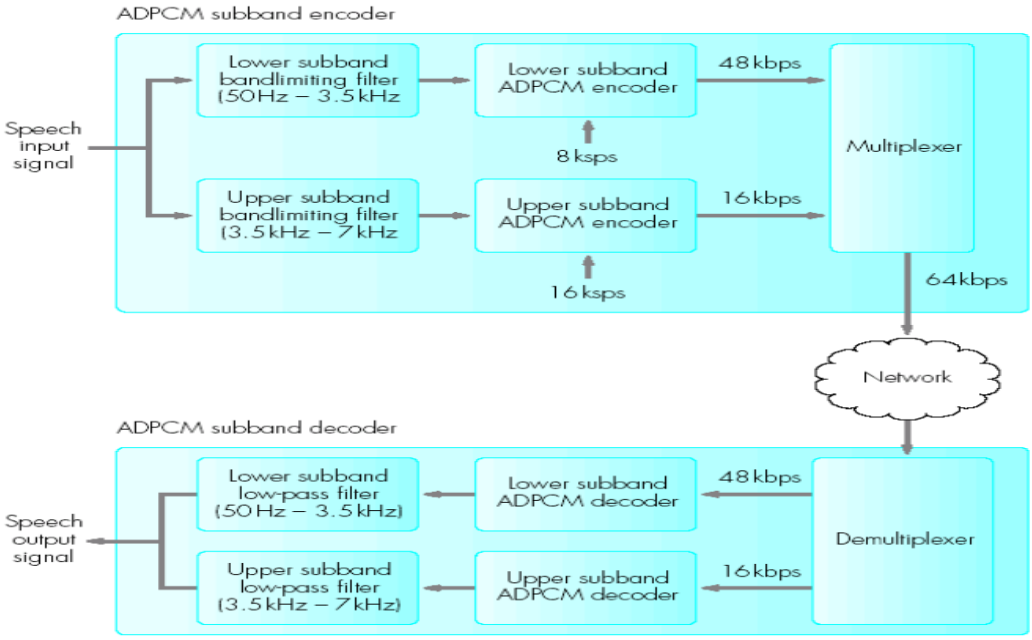
➢ By doing this the input signal is effectively divided into two separate equal-bandwidth signals,

- first known as the lower sub band signal and,
- second known as the upper sub band signal

➢ Each is then sampled and encoded independently using ADPCM.

➢ The use of two sub bands has the advantage that different bit rates can be used for each.

➢ The two bit streams are multiplexed to produce the transmitted signal – in such a way that, decoder in the receiver is able to divide them back again into two separate streams for decoding. Operating bit rates are 64,56, or 48kbps.



Adaptive Predictive Coding:

➢ Even higher levels of compression possible at higher levels of complexity

➢ These can be obtained by also making predictor coefficients adaptive

➢ In practice, the optimum set of predictor coefficients continuously vary since they are a function of the characteristics of the audio signal being digitized

➢ The optimum set of coefficients are then computed and these are used to predict more accurately the previous signal

➢ This type of compression can reduce the bandwidth requirements to 8kbps while still obtaining an acceptable perceived quality

Linear Predictive Coding and Code excited LPC:

Linear predictive coding is a tool used mostly in audio signal processing and speech processing for representing the spectral envelope of a digital signal of speech in compressed form, using the information of a linear predictive model.

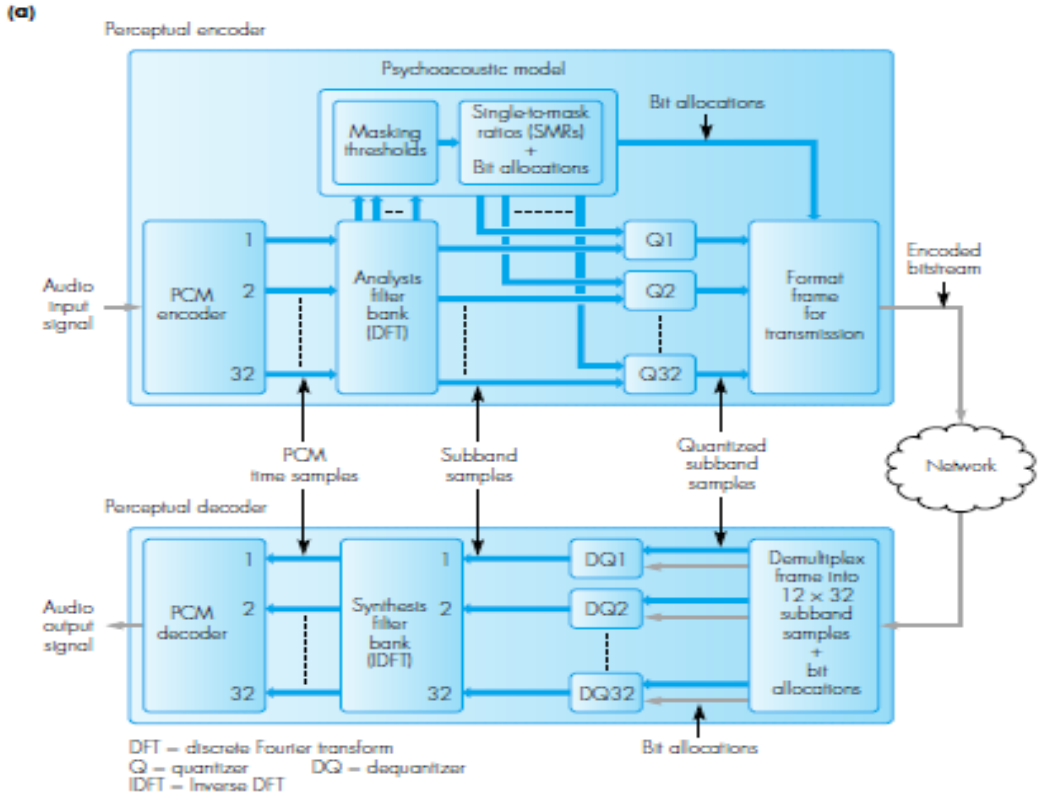
Code Excited Linear Predictive Coding is a speech coding algorithm. CELP coders use codebook excitation, a long term pitch prediction filter, and a short term format prediction filter.

Perceptual Coding:

➢ LPC and CELP are used only for telephony applications and hence compression of speech signals.

➢ PC is designed for compression of general audio such as that associated with a digital television broadcast.

➢ Use a psychoacoustic model (this exploits a number of limitations of human ear).

12.		**MPEG Audio Encoder and Decoder:** 	(10)	CO4	K2

MPEG audio coder

- The audio input signal is first sampled and quantized using PCM.
- The bandwidth available for transmission is divided into a number of frequency sub bands using a bank of analysis filters. Also known as critical-band filters

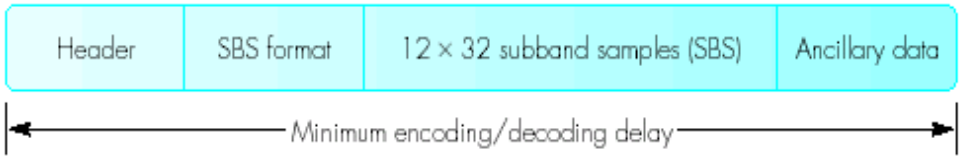
Analysis filter bank:

- Each frequency sub band is of equal width and essentially the bank of filters maps each set of 32 (time related) PCM samples into an equivalent set of 32 frequency samples, one per sub band.
- Hence each is known as a sub band sample and indicates the magnitude of each of the 32 frequency components the are present in a segment of the audio input signal of a time duration equal 32 PCM samples.
- In a basic encoder, the time duration of each sampled segment of the audio input signal is equal to the time to accumulate 12 successive sets of 32 PCM-and hence sub band samples, that is, a time duration equal to 384 (12 x 32) PCM samples.
- The analysis filter bank determines the maximum amplitude of the 12 sub band samples in each sub band. Each is known as the scaling factor for the sub band and these are passed both to the psychoacoustic model and together with the set of frequency samples in each sub band, to the corresponding quantizer block.
- Processing associated with both frequency and temporal masking is carried out by the psychoacoustic model.
- The 12 sets of 32 PCM samples are first transformed into an equivalent set of frequency components using a mathematical technique known as the discrete Fourier transform (DFT)
- Then, using the known hearing thresholds and masking properties of each sub band, the model determines the various masking effects of this set of signals. The output of the model is a set of what are known as signal-to-mask ratios (SMRs) and indicate those frequency components whose amplitude a below the related audible threshold.
- In addition, the set of scaling factors are used to determine the bit allocations, to be used for each of the audible components.
- This is done to have more bits for highest sensitivity regions than those in

regions where the ear is less sensitive.

- In an encoder all the frequency components are carried in a frame.

Frame Format:



HEADER: contains information such as the sampling frequency that has been used

SBS: The peak amplitude level in each sub band is first quantized using 6 bits and a further 4 bits are then used to quantize the 12 frequency components in the sub band relative to this level. Collectively this is called Sub band Sample format.

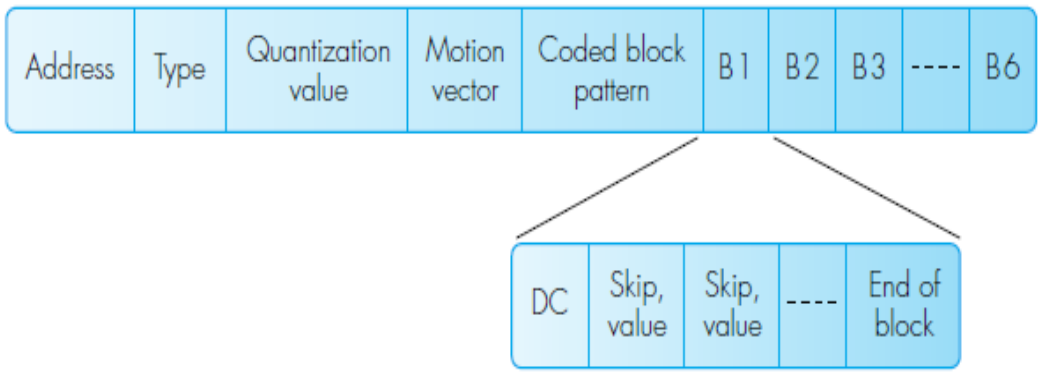
Ancillary data field: at the end of the frame optional. Mainly for future purpose.

13. H.261 video compression standard: (10) CO5 K2

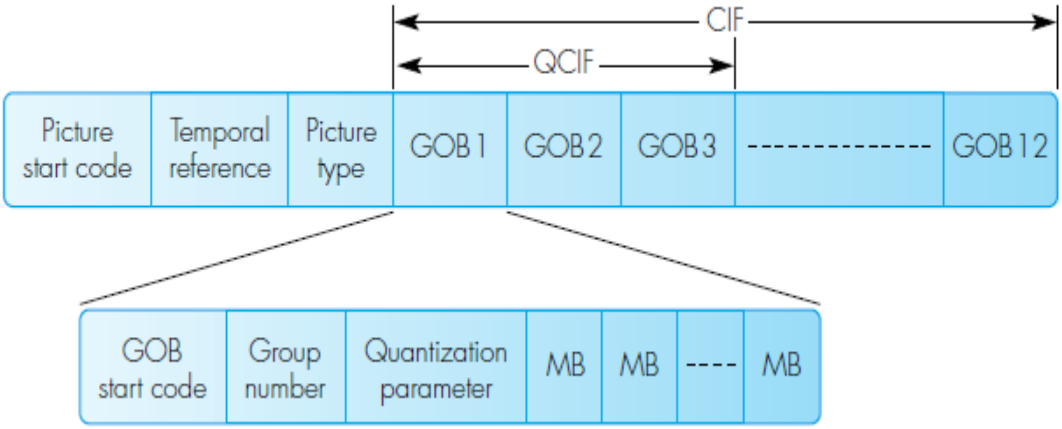
(i) Video formats: CIF and QCIF

Frame Sequence : I P P P I P P P I

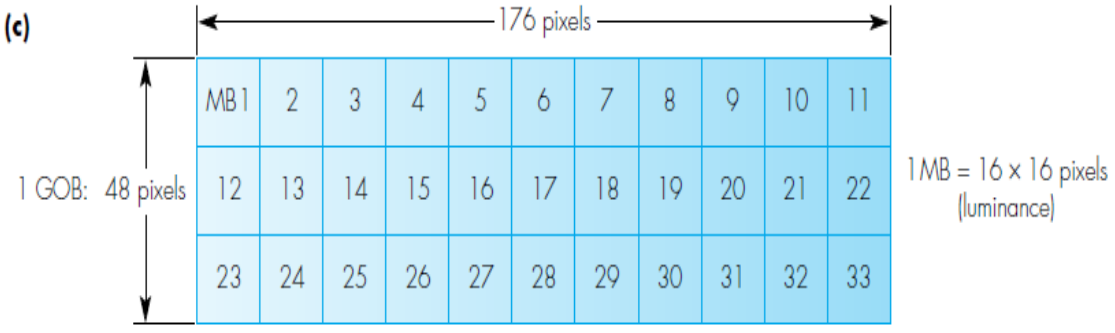
(ii) Macro block Format:



Frame/Picture Format:



(iii) The number of GOBs per frame



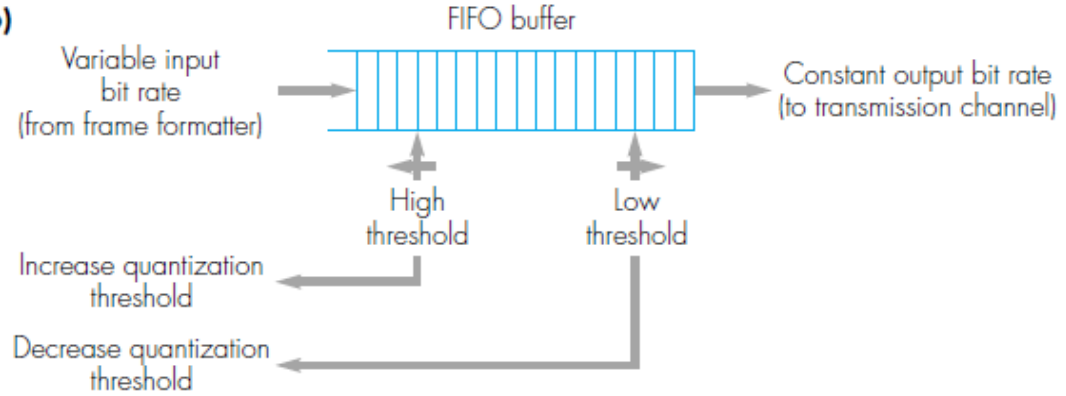
- GOB is matrix of 11 by 3 macro blocks.

- At the head of each GOB is a unique start code which is chosen so that no valid sequence of variable length code words from the table of code words used in the entropy encoding stage can produce the same code.
- In the event of a transmission error affecting a GOB, the decoder simply searches the received bitstream for this code which signals the start of the next GOB.
- For this reason, the start code is also known as the resynchronization marker
- In addition , each GOB has a group number associated with it.

(iv) FIFO buffer

- The transmission bandwidth is available with the H.261 standard is fixed-64 kbps or multiples of 64kbps.
- It is necessary to convert the variable bit rate produced by the basic encoder into a constant bit rate .
- This is achieved by first passing the encoded bitstream output by the encoder through a first-in, first-out (FIFO) buffer prior to it being transmitted and then providing a feedback path from this to the quantizer unit within the encoder.

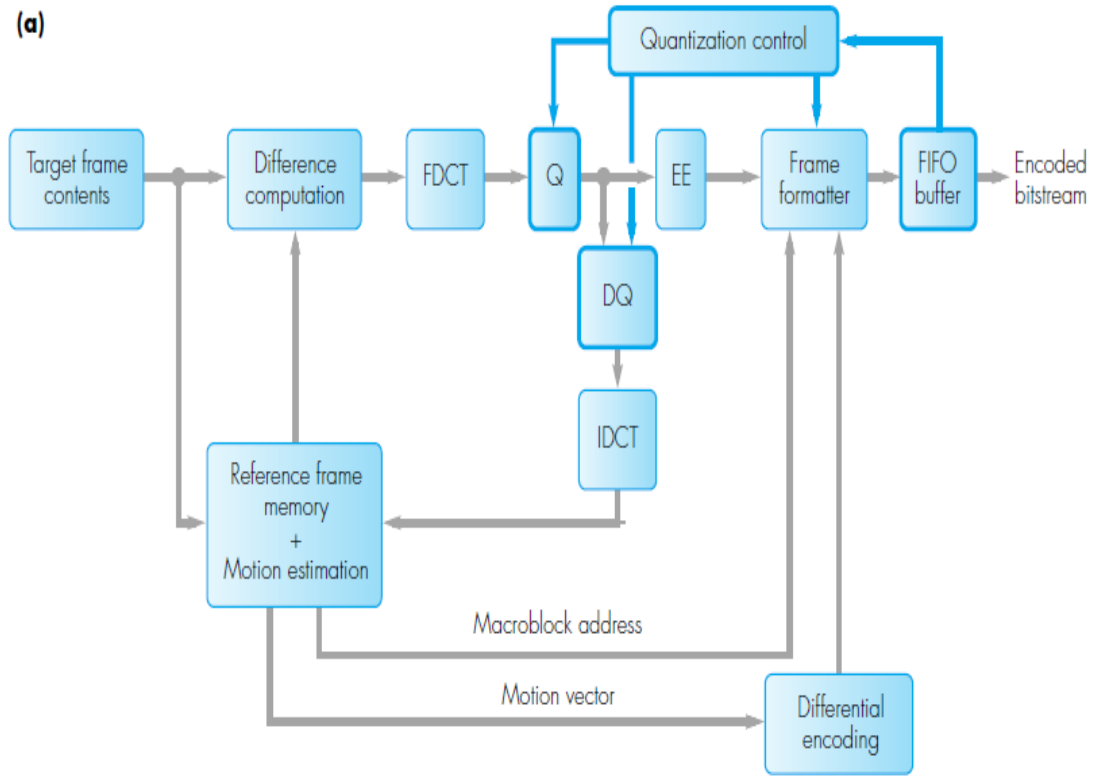
(b)



- To obtain a constant output bit rate from the encoder by dynamically varying the quantization threshold used. This is the role of the FIFO buffer.
- The output rate from the buffer is constant-determined by the (constant) bit rate of the transmission channel
- If the input rate temporarily exceeds the output rate, then the buffer will start to fill. Conversely, if the input rate falls below the output rate, then the buffer contents will decrease.

(v)Video encoder

(a)



14.	i)	i)M is the distance (in frames) between a P-frame and the immediately preceding I- or P- frame, and N is the number of frames between two successive I-frames and known as group of pictures or GOP. Hence for the frame sequence shown in Figure is M = 3 and N = 12. ii)A suitable reordered frame sequence that meets the defined requirements is: IPBBPBBPBBIBBPBB	(4)	CO5	K3
	ii)	Frame sequence = IBBPBBPBBPBBI... Hence: 1/12 of frames are I-frames, 3/12 are P-frames, and 8/12 are B-frames. **Average compression ratio** = (1 × 0.1 + 3 × 0.05 + 8 × 0.02)/12 = 0.0342 or 29.24: 1 **NTSC frame size:** Without compression= 352 × 240 × 8 + 2 (176 × 120 × 8) = 1.013760Mbits per frame With compression = 1.01376 × 1/29.24 = 34.670kbits per frame Hence bit rate generated at 30fps = 1.040Mbps **PAL frame size:** Without compression= 352 × 288 × 8 + 2 (176 × 144 × 8) = 1.216512Mbits per frame With compression = 1.216512 × 1/29.24 = 41.604kbits per frame Hence bit rate generated at 25fps = 1.040Mbps	(6)	CO5	K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	33.3	50	16.7	--	--	--